Introduction to Unity – Game Development Basics

Software Engineering e-portfolio presentation

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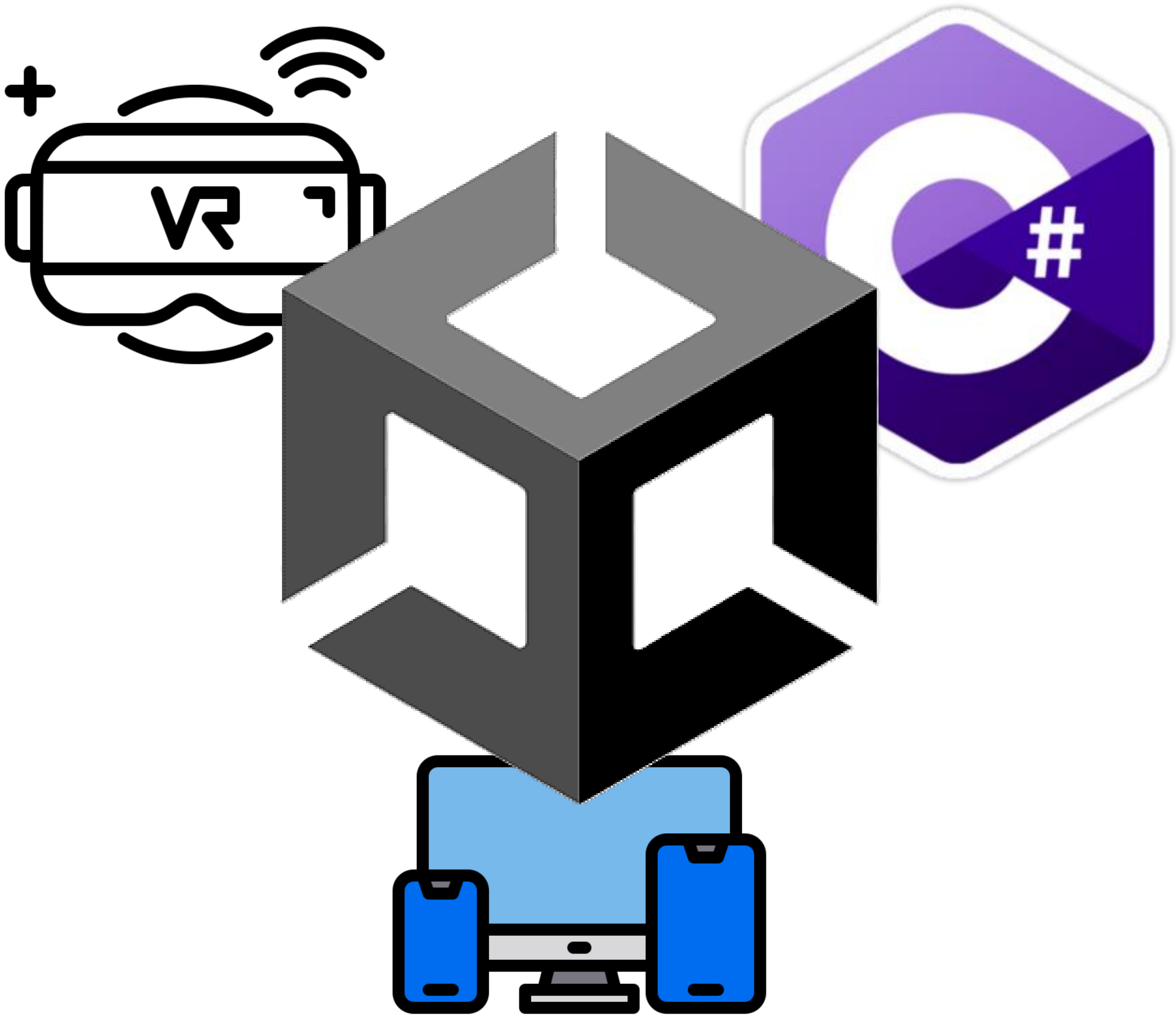
New Study 2024, 4th Semester
7th May 2026

Agenda

- What is Unity?
- Why / Where do we use Unity?
- Core Concepts
- Use Cases & Examples
- Alternatives
- Live Demo
- Extra Showcase



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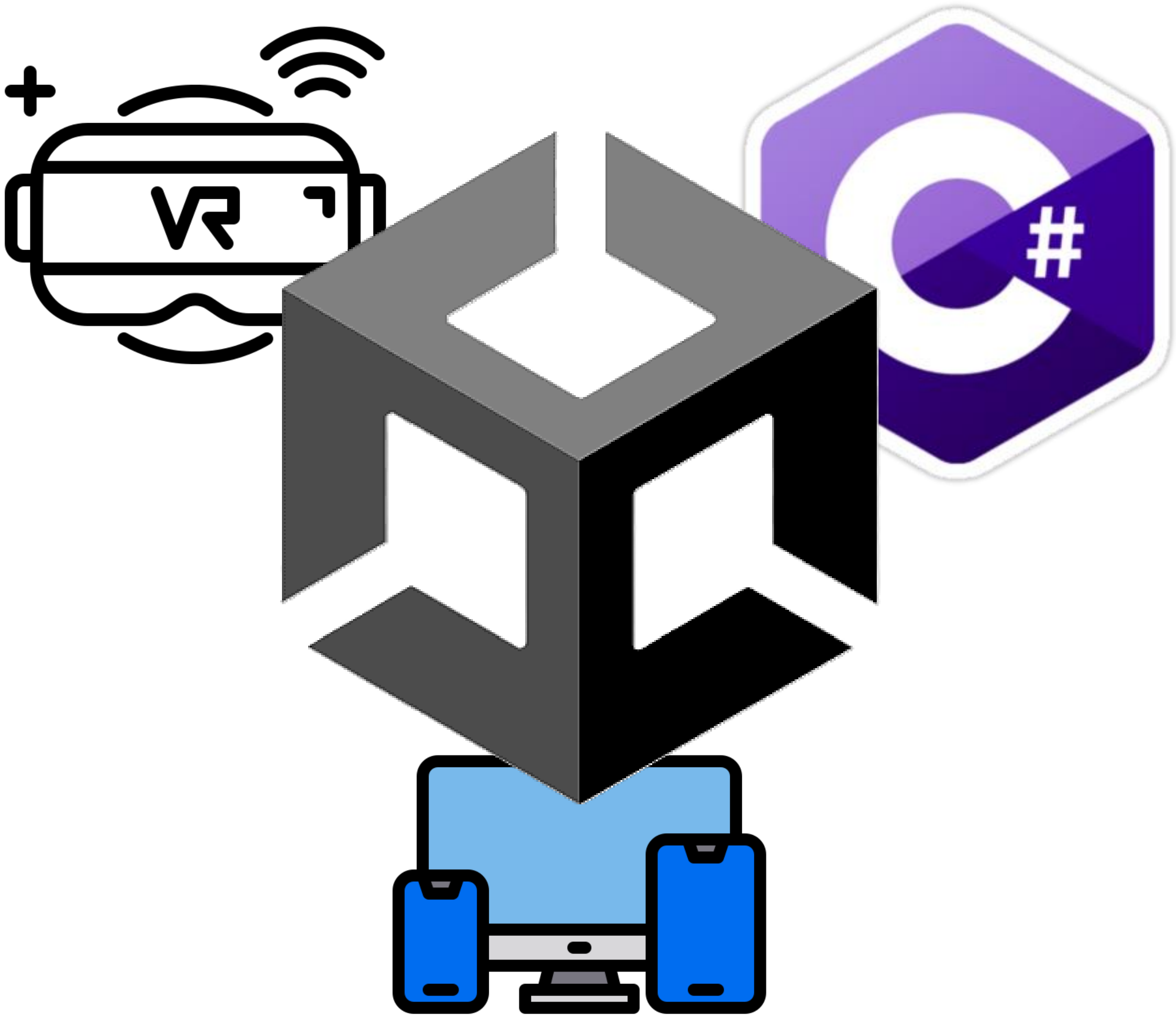
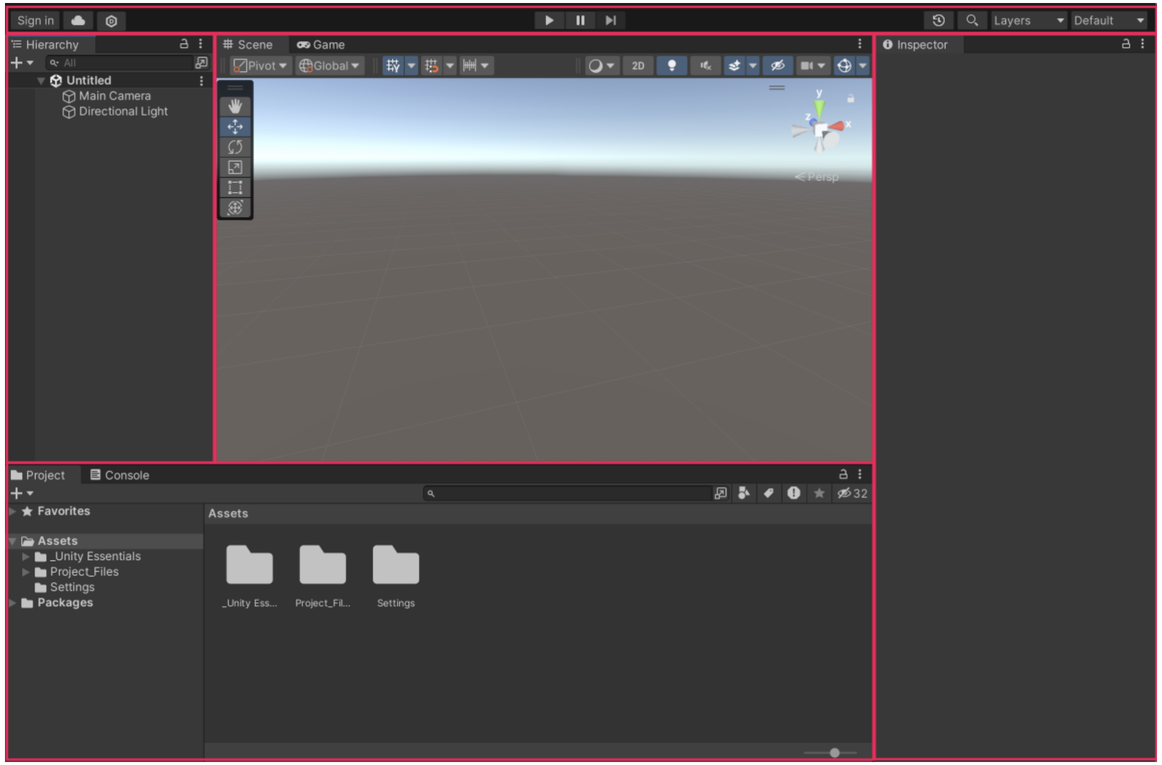


What is Unity?

→ Unity is a real-time development platform for creating interactive 2D and 3D applications

Key features:

- Visual scene editing
- C# scripting
- Built-in physics, audio, input and rendering
- Multi-platform deployment



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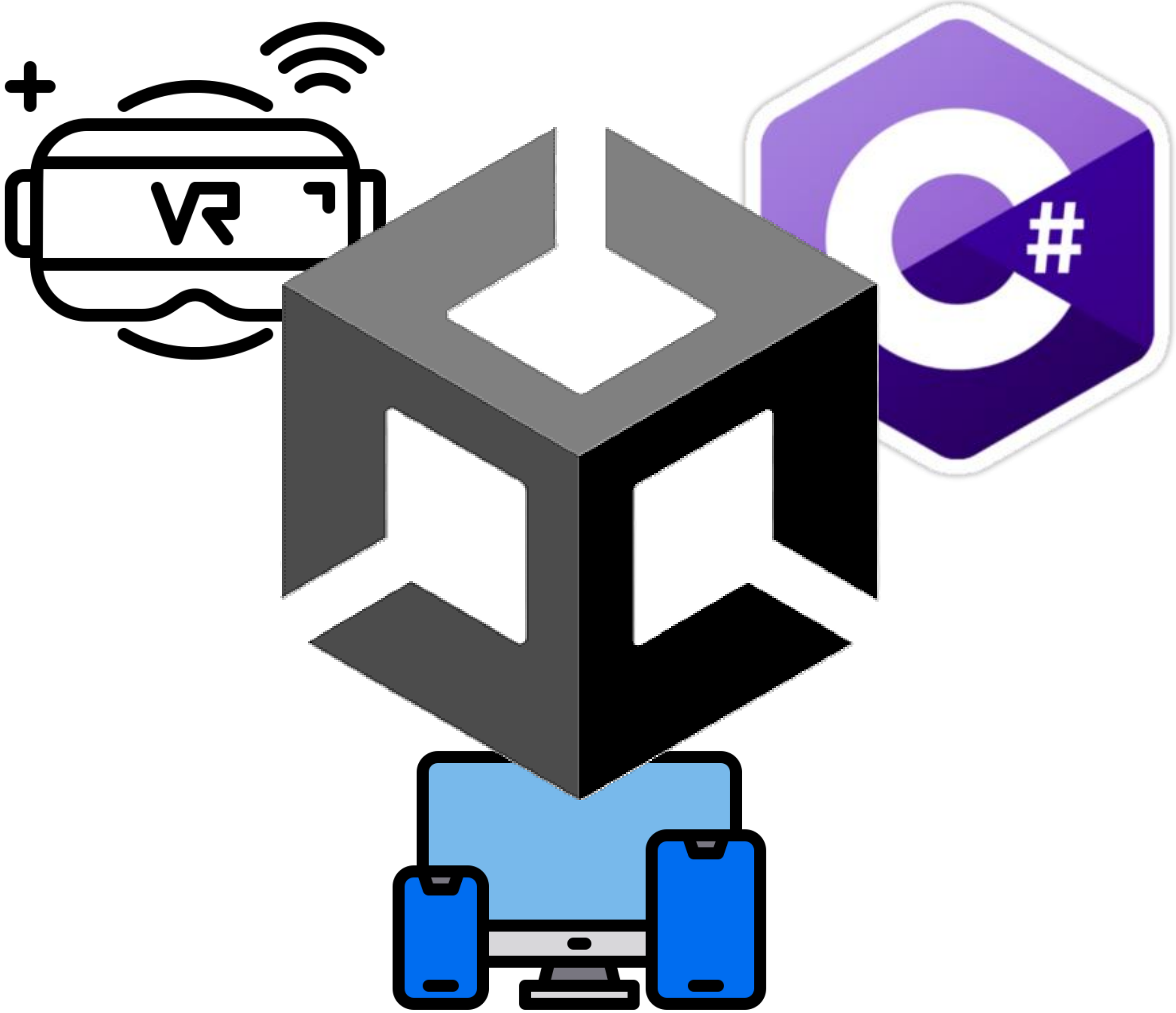

Why do we use Unity?

- Easy to learn
- C# scripting
- Cross-platform
- Large community
- Asset Store

Where do we use Unity?

- Mobile & indie games
- VR / AR applications
- Simulations & training
- Architecture & visualization
- Film & animation

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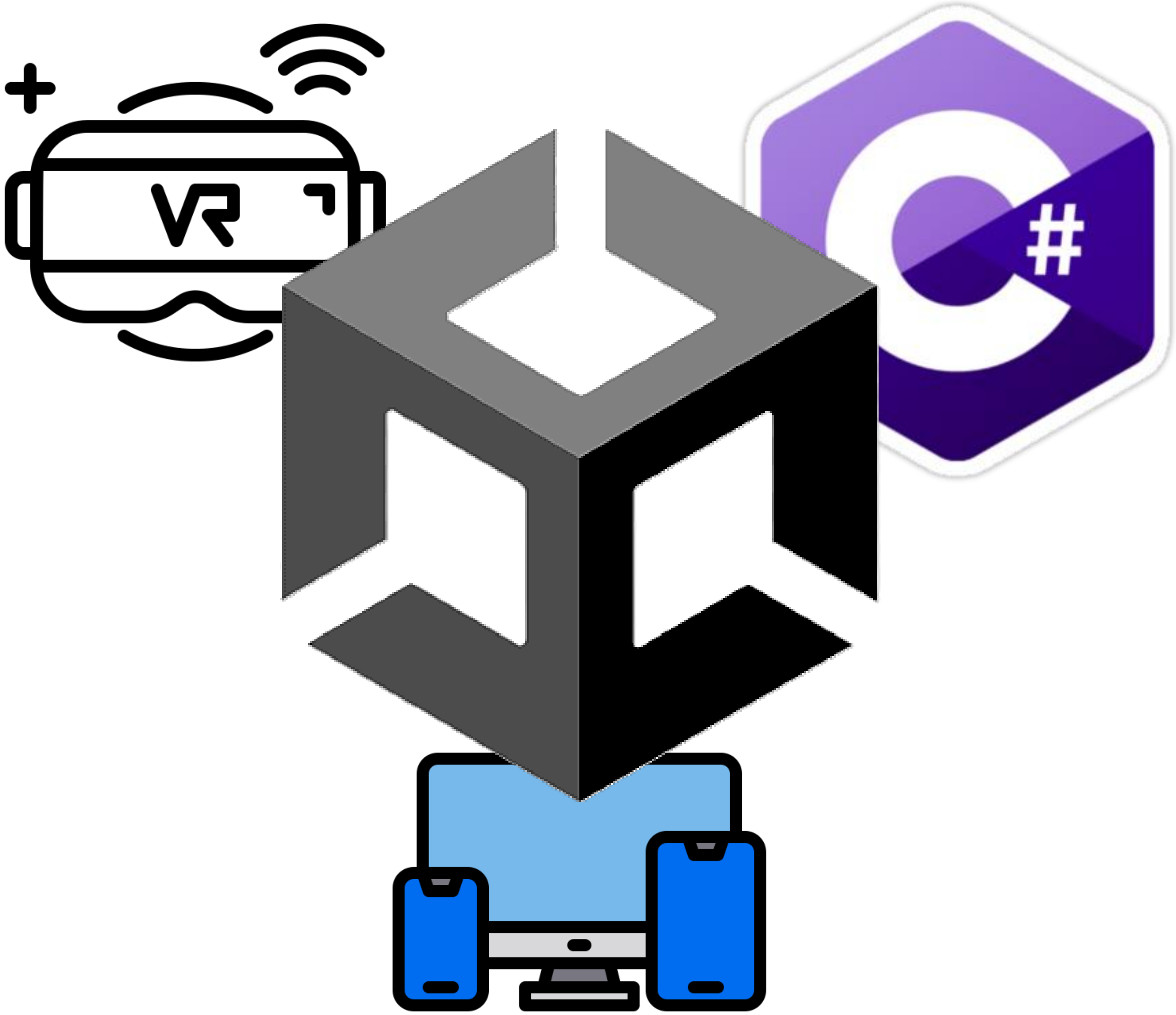


Core Concepts

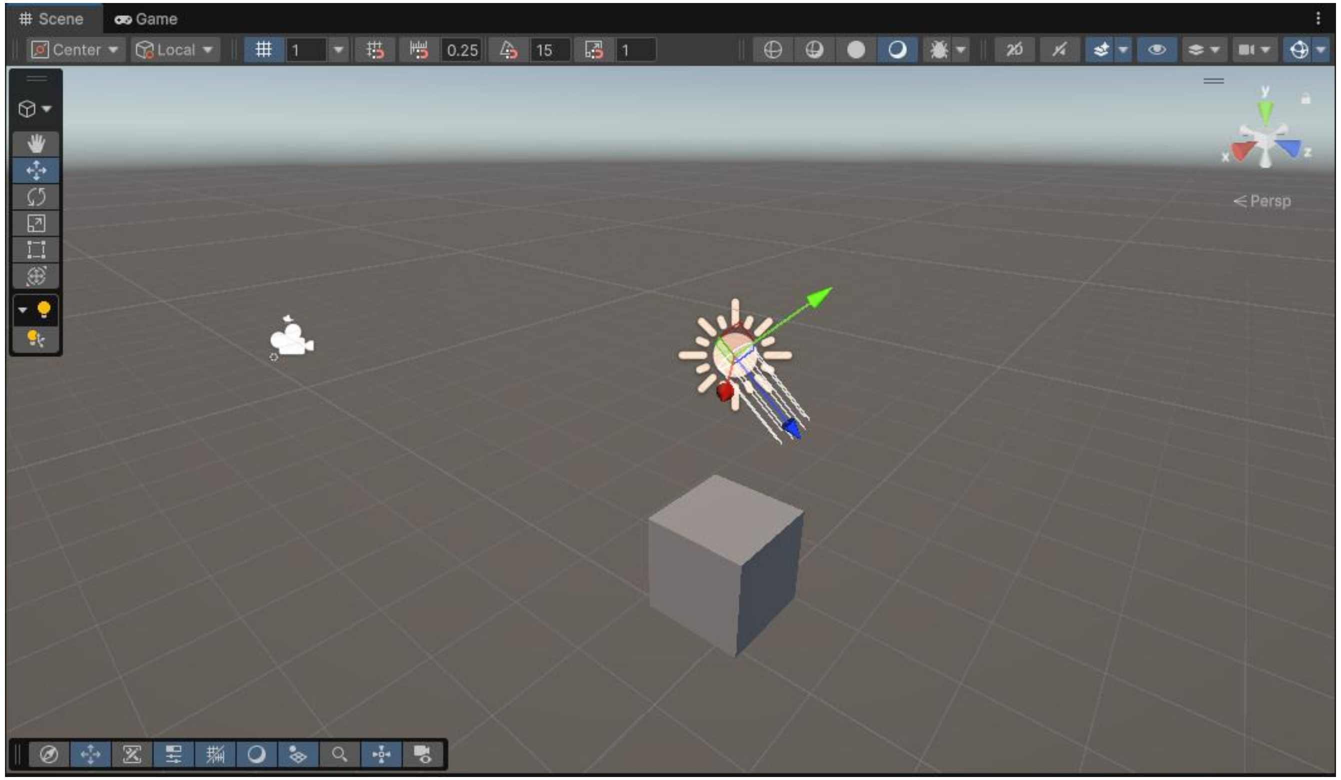
→ Everything in Unity is a GameObject inside a Scene (your level)

A Scene:

- Contains all GameObjects
- Represents a level

GameObjects can be:

- Cube
- Camera
- Source of light
- …



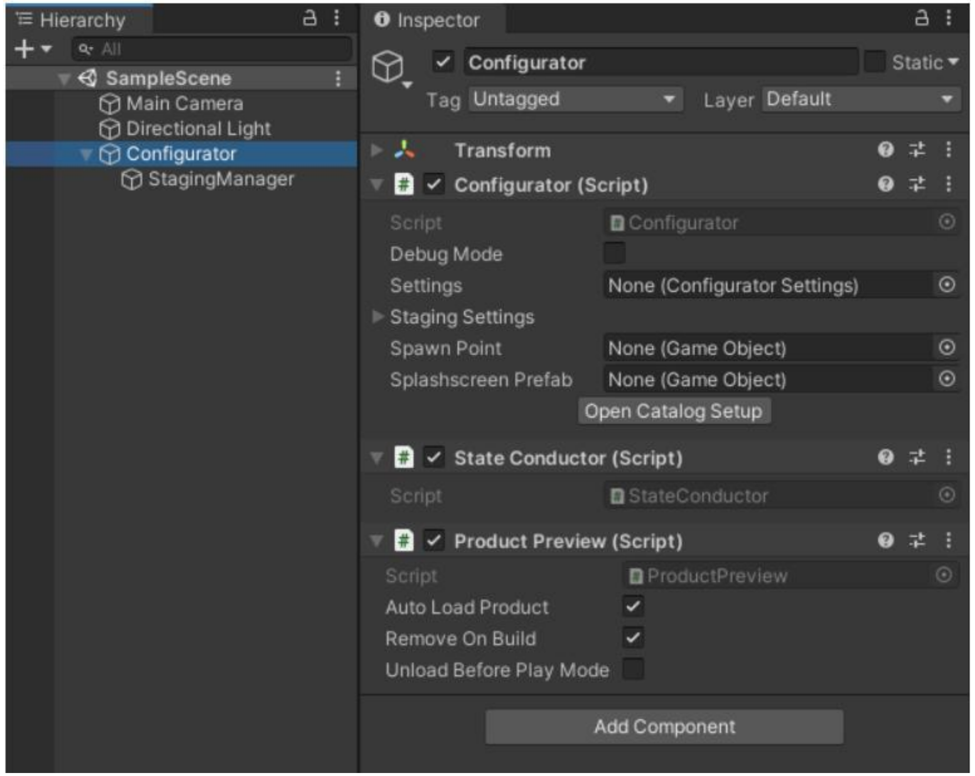
Core Concepts

→ Components define the behavior and properties of a GameObject

Examples for Components are:

- Transform (position, rotation, scale)
- Renderer (appearance)
- Collider (physics)

Components give GameObjects their functionality!



Core Concepts

→ Scripts allow you to create your own logic and attach them to GameObjects

- Add custom behavior
- Written in C#
- Attached as components

```csharp
using UnityEngine;

public class TestScript : MonoBehaviour
{
    // Start is called once before the first execution of Update after the MonoBehaviour is created
    void Start()
    {

    }

    // Update is called once per frame
    void Update()
    {
        transform.Rotate(0, 50 * Time.deltaTime, 0);
    }
}
```

What does this Script do?

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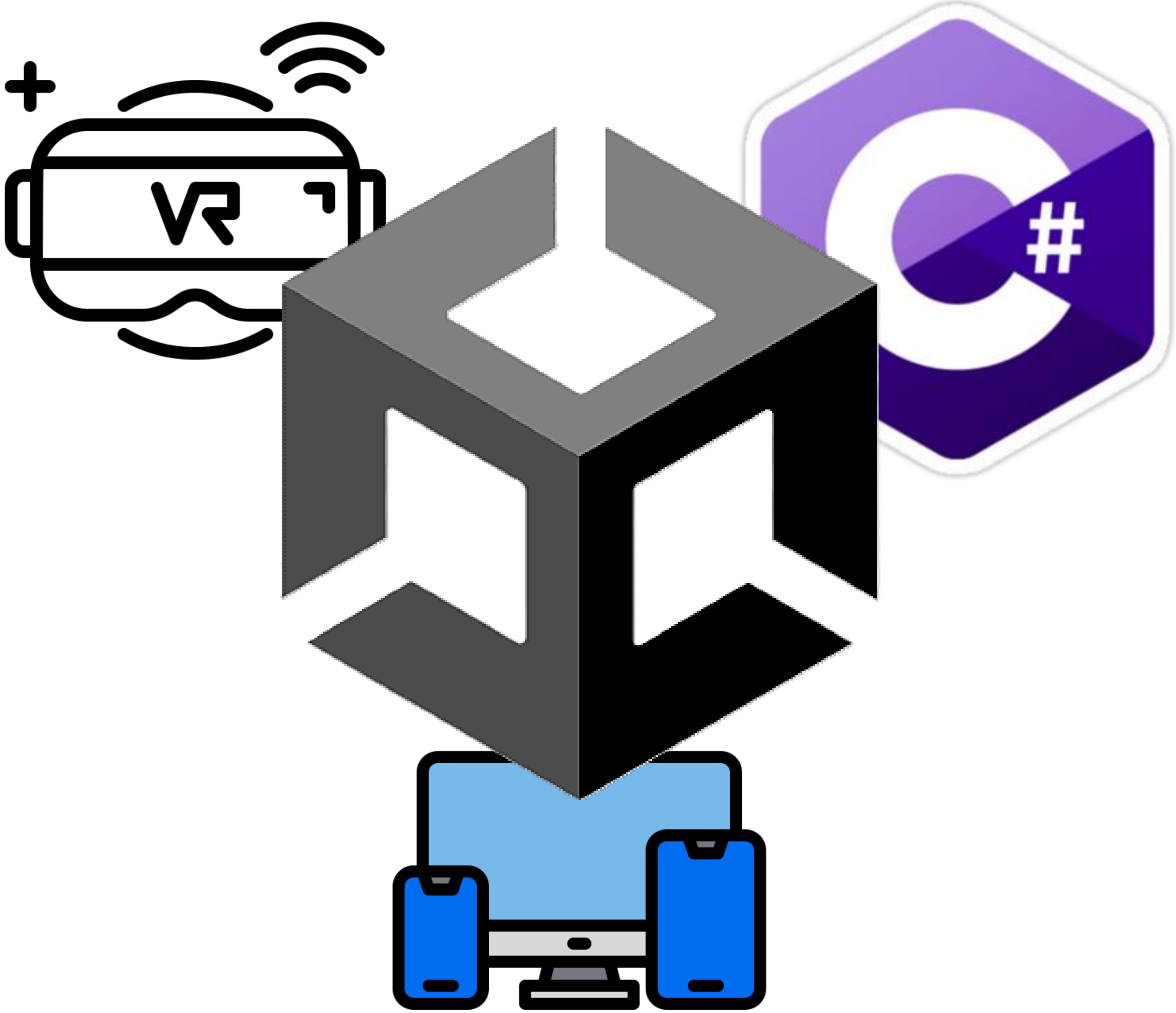

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Use Cases & Examples

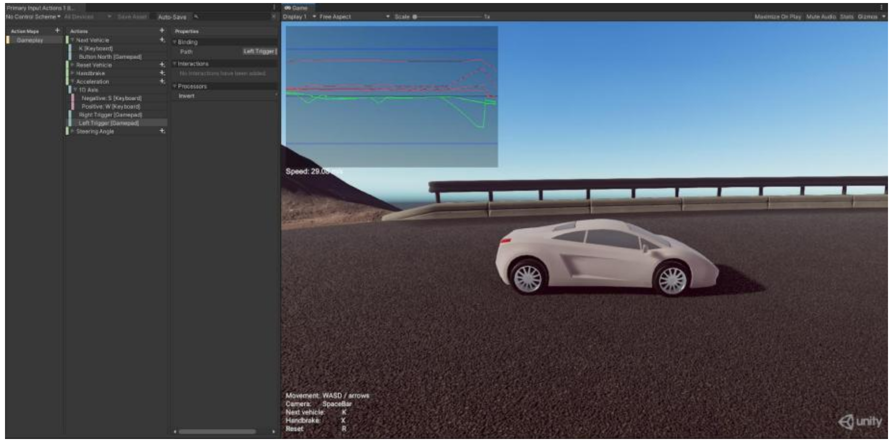
1. Games

- Among Us
- Pokémon GO
- Cuphead
- Hollow Knight

Use Cases & Examples

2. Beyond Games

VR / AR



Automotive / Simulation

Architecture

Medical Training

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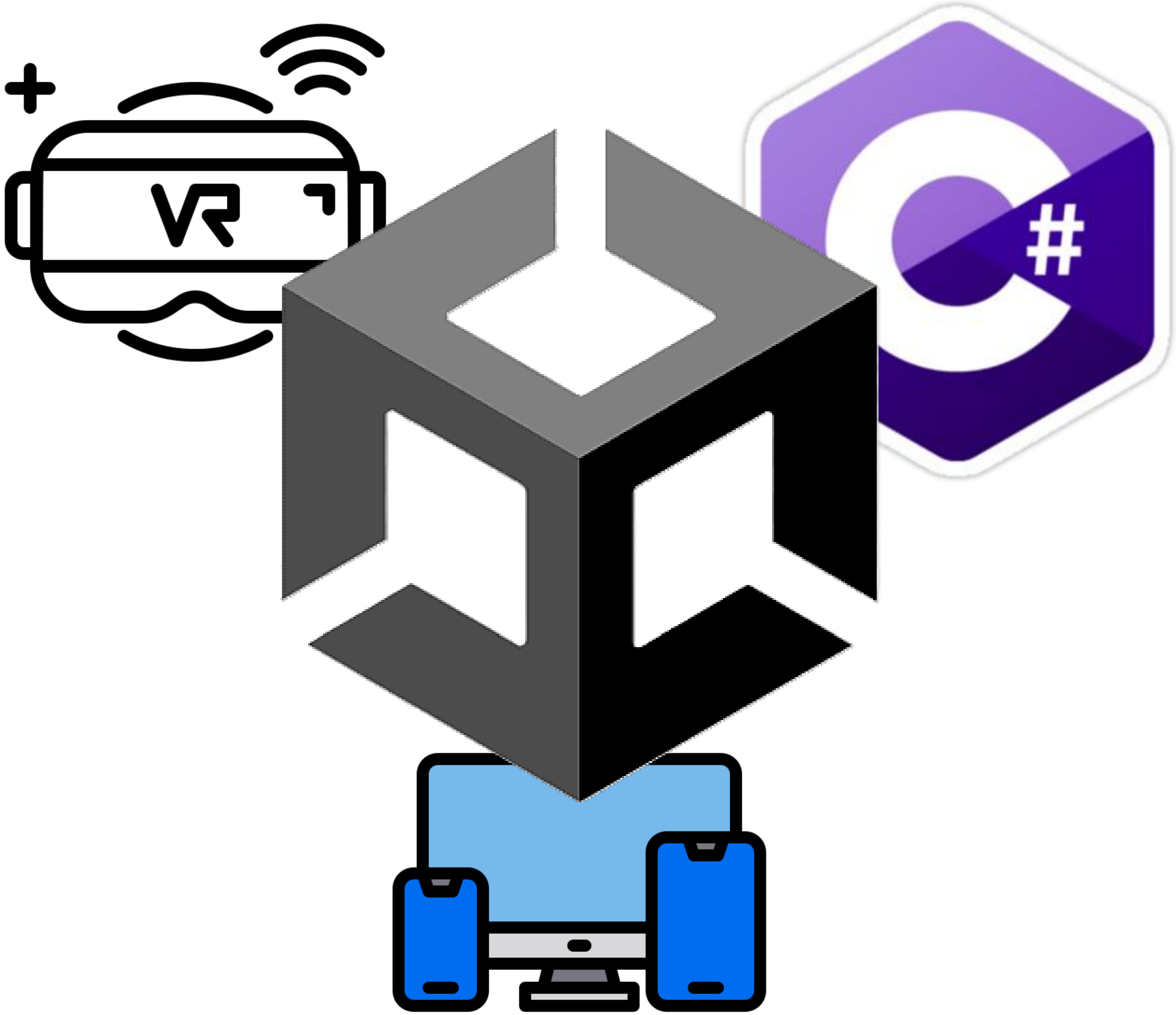
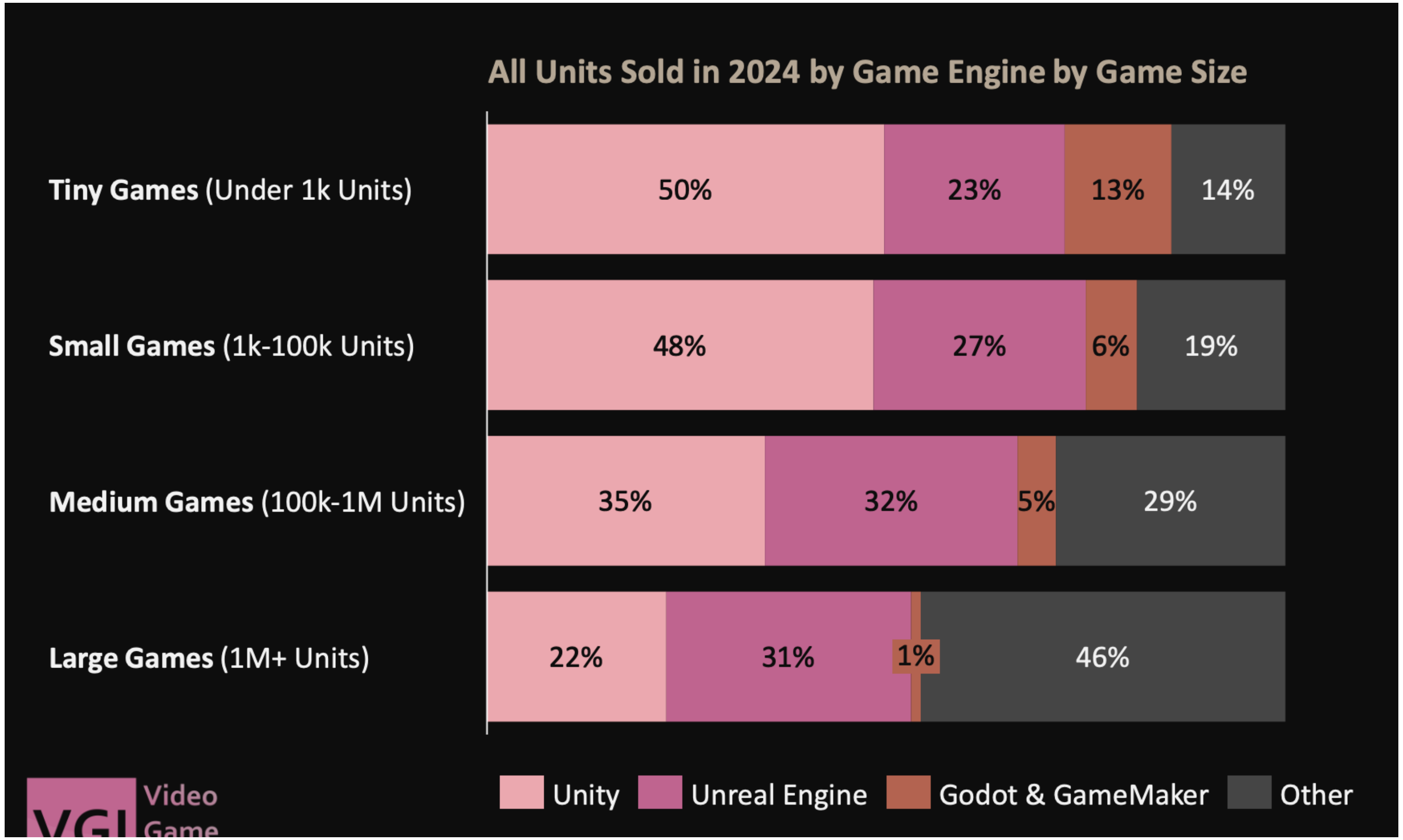


Alternatives

Other popular engines are:

- Unreal Engine
- Godot
- CryEngine
- GameMaker



Alternatives

Feature	Unity	Unreal	Godot
Ease of Use	Easy to learn, beginner-friendly	More complex, steeper learning curve	Easy, lightweight, very beginner-friendly
Graphics	Good, flexible (not top-tier by default)	Excellent, industry-leading (AAA quality)	Basic to moderate (improving, but limited)
Language	C#	C++ / Blueprints	GDScript
Performance	Good (depends on optimization)	Very high (best for high-end games)	Good for small to medium projects
Use Case	Indie & mobile games, VR / AR applications, Simulations & training, Architecture & visualization, Cross-platform apps	AAA Games, Film & virtual production, High-end simulations, Automotive visualization	Indie games, 2D games, Hobby & learning, Open-source projects, Lightweight applications

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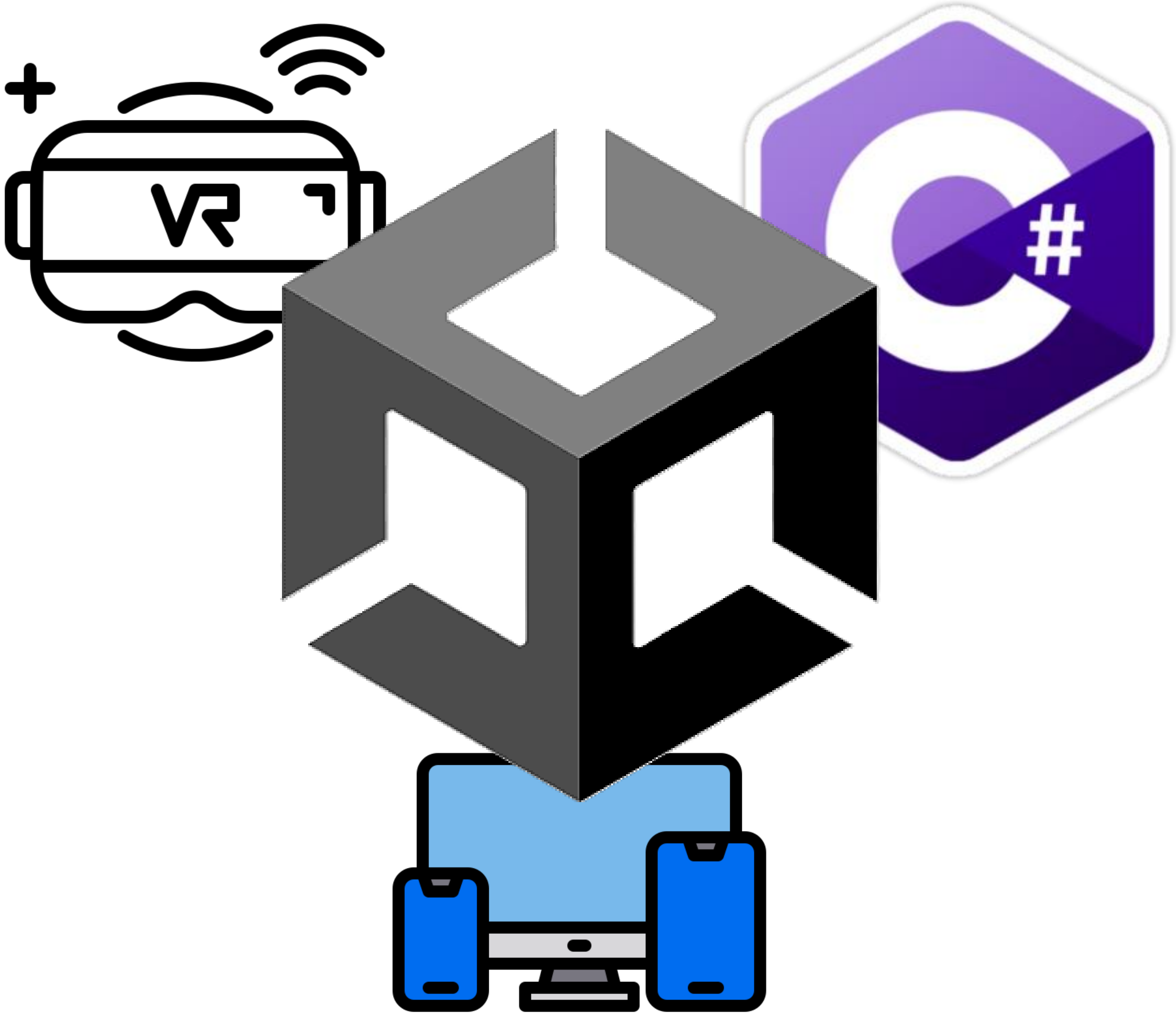

Live Demo

→ Let’s build our first 3D scene together!

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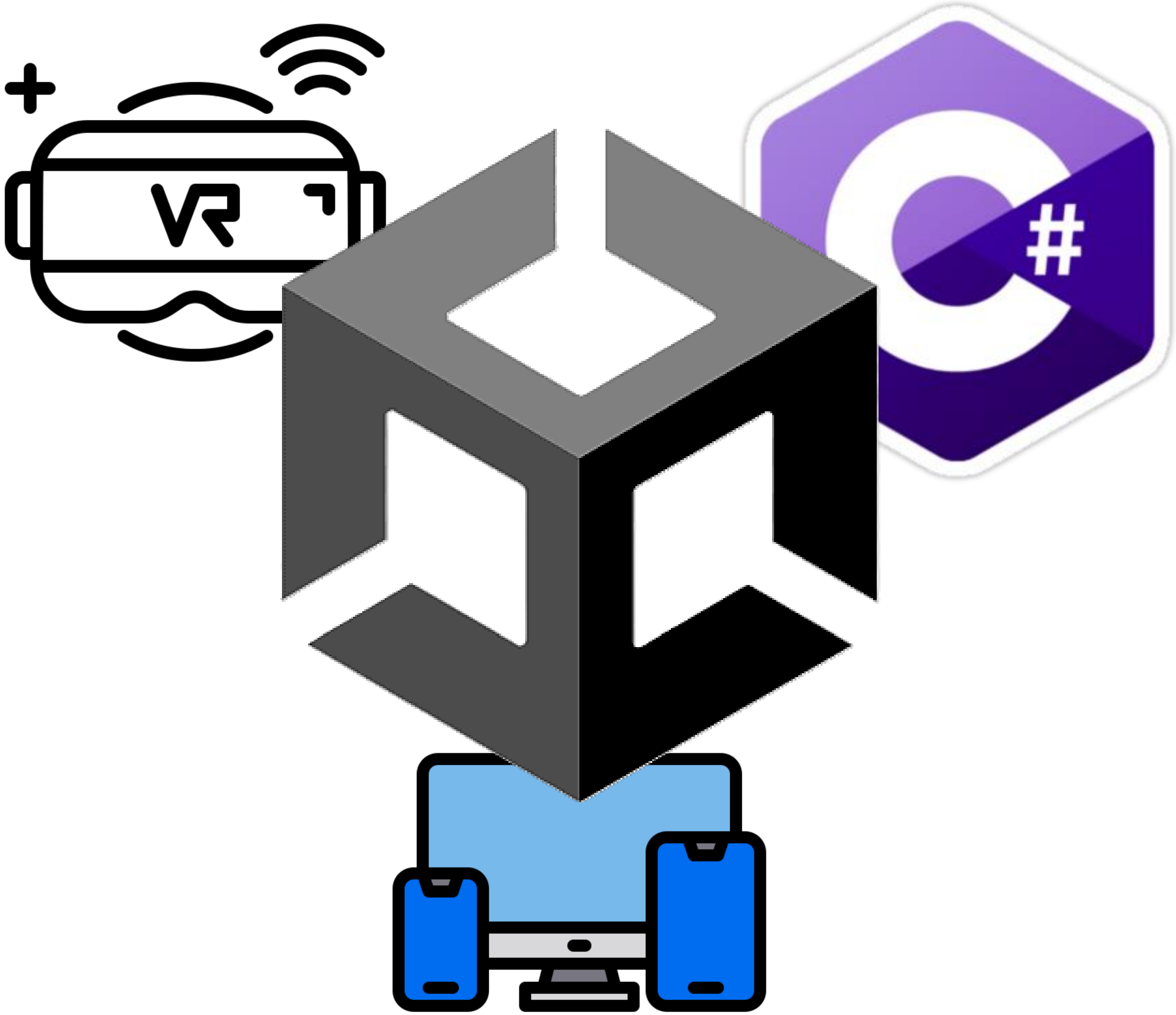
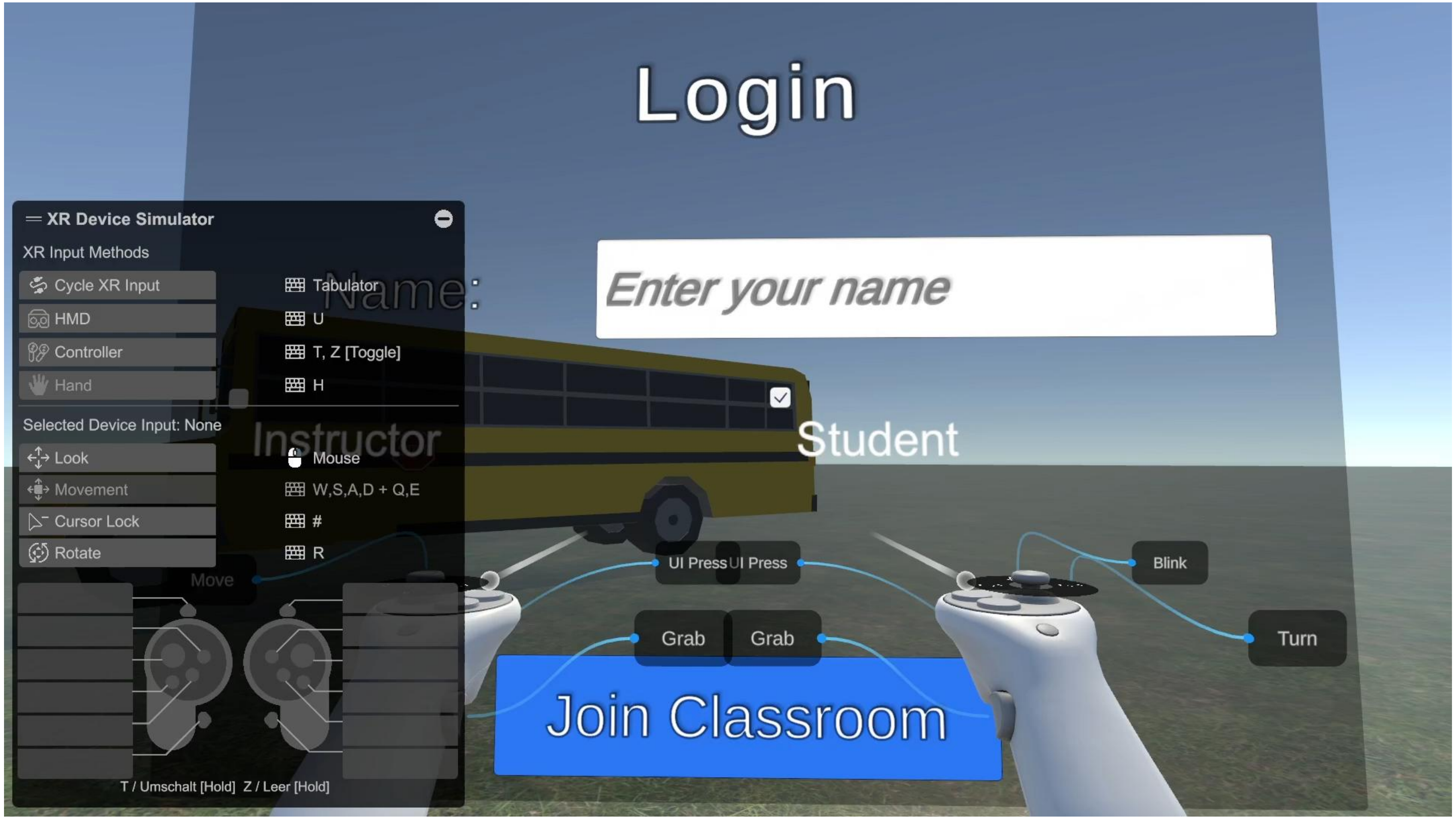


Extra Showcase



Thank you for your attention!

Any questions?